

AI-ASSISTED LEGAL VOICE ASSISTANT WITH SPEECH-TO-TEXT AND RVC-BASED VOICE CONVERSION

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1 Introduction and Background

1.1 Problem Description

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have enabled the development of domain-specific intelligent assistants capable of answering user queries based on internal knowledge sources. However, most existing systems remain limited to text-based interaction, which reduces usability and accessibility.

Voice-based interaction has been widely studied as a more efficient alternative. Speech production typically reaches 120–150 words per minute, compared to approximately 40 words per minute for typing. This enables faster information input and improves user experience, especially for long-form queries.

Despite these advantages, existing Text-to-Speech (TTS) systems often produce synthetic and less expressive voices. This creates a gap between efficient input (speech) and natural output.

Therefore, this research proposes to extend an existing RAG-based legal assistant by integrating a bidirectional voice interface, including Automatic Speech Recognition (ASR) for speech input and Retrieval-based Voice Conversion (RVC) for personalized speech output.

1.2 Research Objectives

The objectives of this project are:

- Integrate a speech-to-text module (Whisper) into an existing RAG-based legal assistant.
- Implement a TTS + RVC pipeline for personalized voice output.
- Evaluate system performance in terms of latency, transcription accuracy, and voice quality.
- Compare the proposed approach with zero-shot voice cloning methods (XTTS-v2).

1.3 Scope of the Project

This project focuses on integrating voice capabilities into an existing AI assistant system. It does not involve developing new neural network architectures. The implementation is limited to a prototype system supporting Vietnamese and English.

2 Literature Review

Voice-based interaction has been extensively studied in Human-Computer Interaction (HCI). Prior research shows that speech input can significantly improve text entry speed compared to typing, particularly for long-form content.

Modern AI systems increasingly integrate speech technologies, including Automatic Speech Recognition (ASR) and neural Text-to-Speech (TTS). While TTS models such as Tacotron produce intelligible speech, they often lack naturalness and speaker-specific characteristics.

Retrieval-based Voice Conversion (RVC) has emerged as a promising solution for personalized speech generation. RVC transforms source audio into a target speaker’s voice while preserving linguistic content and prosody. Compared to zero-shot voice cloning models such as XTTS-v2, RVC typically provides higher speaker fidelity at the cost of additional training and system complexity.

2.1 Comparison of Approaches

Criteria	TTS	TTS + RVC	XTTS-v2
Naturalness	Synthetic	Natural	Natural
Personalization	Low	High	Medium
Training Cost	None	Moderate	None
Latency	Low	Medium	Medium

Table 1: Comparison of voice generation approaches

3 Proposed Solution

3.1 System Overview

The proposed system extends an existing RAG-based assistant with a voice interaction pipeline consisting of:

- Audio input (user speech)
- ASR (Whisper)
- RAG retrieval + LLM (Llama 3.1)
- TTS (edge-TTS)
- RVC voice conversion
- Audio output

RAG retrieval provides relevant legal documents as context, improving the factual accuracy of generated responses.

3.2 High-level Pipeline

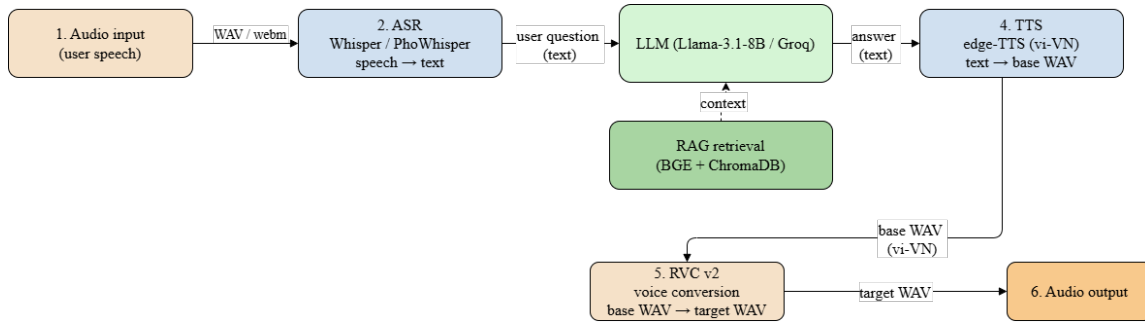


Figure 1: High-level abstract pipeline of the Legal Voice Assistant

3.3 Software Architecture

The system follows a client-server architecture. The backend handles ASR, RAG processing, and TTS generation, while RVC inference is deployed on a separate GPU environment using Google Colab.

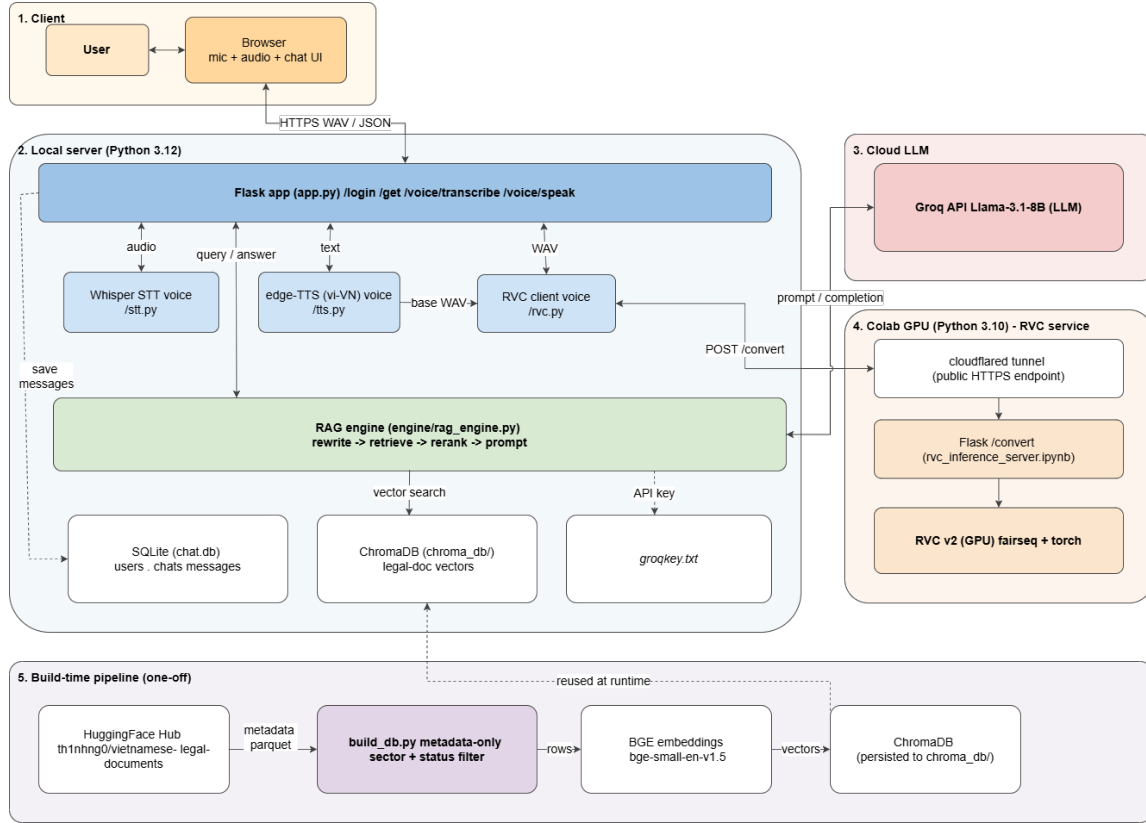


Figure 2: Software architecture of the system

3.4 Technology Stack

- ASR: OpenAI Whisper / PhoWhisper
- LLM: Llama-3.1 (Groq API)
- RAG: BGE embeddings + ChromaDB
- TTS: edge-TTS (Vietnamese voices)
- Voice Conversion: RVC v2
- Backend: Python + Flask

4 Implementation Plan

4.1 Development Stages

- Dataset preparation
- RVC model training (Google Colab)
- Backend integration

- Frontend integration
- Evaluation and testing

4.2 Schedule

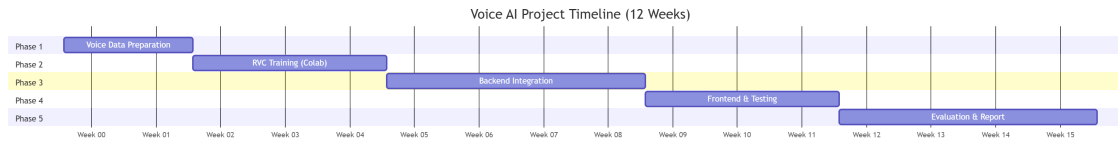


Figure 3: Project timeline (12 weeks)

4.3 Feasibility

The project is feasible due to the availability of open-source tools and cloud-based GPU resources. Potential challenges include latency and audio quality, which are addressed through system optimization techniques.

5 Expected Results

- A working prototype with voice interaction
- Trained RVC voice models
- Performance evaluation report
- Thesis document and demo video

6 Evaluation Plan

The system will be evaluated using both objective and subjective metrics:

- Word Error Rate (WER)
- End-to-end latency
- Signal-to-Noise Ratio (SNR)
- Mean Opinion Score (MOS)

7 References

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4. Y. Wang et al., "Tacotron: Towards End-to-End Speech Synthesis," 2017.